MIL-I-27273A
3 JUNE 1965

Superseding
MIL-I-27273(USAF)
5 November 1959

MILITARY SPECIFICATION

INVERTER, POWER, STATIC, GENERAL SPECIFICATION FOR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the general requirements for static inverters for use in converting electrical power in flight vehicles from 28V dc to ac power at 400 cps.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electrodeposited)

Military

MIL-P-116	Preservation, Methods of
MIL-M-3171	Magnesium Alloy, Processes for Corrosion Protection of
MIL-C-3965	Capacitors, Fixed, Nonsolid Electrolytic (Tantalum, Foil, and Sintered-Slug) General Specification for
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-S-7742	Screw Threads, Standard, Aeronautical
MIL-P-7936	Parts and Equipment, Aeronautical, Preparation for Delivery
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-E-26144	Electric Power, Missile, Characteristics and Utilization, General Specification for
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MIL-STD-704	Characteristics and Utilization of Aircraft Electric Power

FSC 6125

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:29Z
Check the source to verify that this is the current version before use.

MIL-STD-826	Electromagnetic Interference Test Requirements and Test Methods
MIL-STD-831	Test Reports, Preparation of
MS33543	Criteria - Temperature and Altitude Range Self Cooled Electric Equipment
MS33586	Metals, Definition of Dissimilar

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification.- The inverter furnished under this specification shall be a product that has been tested and passed the qualification tests herein and has been listed on or approved for listing on the applicable qualified products list.

3.2 Data.- Unless otherwise specified in the contract or order, no data (other than reports and drawings accompanying qualification samples) are required by this specification or any of the documents listed in section 2 (see 6.2).

3.3 Selection of specifications and standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143, except as provided in 3.3.1 and 3.3.2.

3.3.1 Military standard parts.- Military standard parts (MS and AN) shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their part numbers. In the event there is no suitable corresponding standard part in effect on date of invitation for bids, commercial parts may be used, provided they conform to all requirements of this specification.

3.3.2 Commercial parts.- Commercial utility parts, such as screws, bolts, nuts, and cotter pins may be used, provided they possess suitable properties and are replaceable by the MS or AN parts without alteration, and provided the corresponding MS or AN part numbers are referenced in the parts list and, if practicable, on the contractor's drawings.

3.4 Materials.-

3.4.1 Fungus-proof materials.- Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they will be used in a hermetically sealed enclosure, fungicidal treatment will not be necessary.

3.4.2 Metals.- Metals shall be corrosion resistant or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions that may be encountered in storage or normal service.

3.4.2.1 Dissimilar metals.- Unless suitably protected against electrolytic corrosion, dissimilar metals as defined in MS33586 shall not be used in intimate contact with each other.

Source: https://assist.dla.mil -- Downloaded: 20260-20T11:29Z
Check the source to verify that this is the current version before use.

...w after ... cluded in the contract or order (see 6.1.2) and submitted to NAVSEC for rev...

3.5 Design and construction.- The inverter shall be designed and constructed in accordance with this specification and the applicable MS.

3.5.1 Serviceability.- The inverter shall be designed and constructed in such a manner that no parts will work loose in service and shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.2 Unless otherwise specified on the applicable MS, the inverter shall be a completely static unit having no moving parts such as relays or contactors. However, a tuning fork may be used as a frequency reference provided its use will not cause the inverter to malfunction under conditions of shock and vibration.

3.5.3 Reliability.- The inverter shall be designed and constructed as specified herein in a manner that will insure the highest degree of operational reliability under all service conditions. The inverter shall have a minimum number of parts consistent with the requirements specified herein. The specified performance and characteristics of the equipment shall not be changed to achieve the reliability requirements without the specific approval of the procuring activity.

3.5.4 Input.- The rated input voltage shall be as specified on the applicable MS. The inverter shall be designed to operate with no evidence of instability from a flight vehicle system delivering power conforming to MIL-STD-704 and MIL-E-26144 within the voltage limits specified on the applicable MS.

3.5.4.1 Influence on electrical system.- The influence on the power supply system shall not exceed the limits specified in the paragraph titled "Influence on electric system" of MIL-STD-704.

3.5.6 Electrical connections.- The input and output connections shall be in accordance with the connections shown on the applicable MS. The design shall be such that no internal connections will be required to obtain 3-phase wye, 3-phase delta, and single-phase output.

3.5.7 Safety wiring and staking.- Accidental loosening of screws, screw parts, and other connections shall be prevented by safety wiring, where practicable (0.32-inch minimum OD), staking, or other approved methods. Washers and cotter pins, where used, shall be assembled in a manner to prevent rotation of washers and movement of cotter pins under conditions of vibration.

3.5.8 Ventilation openings.- All ventilation openings shall be designed to prevent passage of foreign objects. When all covers are in place, vulnerability of the inverter to damage from short circuits caused by loose metallic objects shall be reduced to a minimum.

3.5.9 Adjustment.- The inverter shall require no external adjustment for voltage or frequency regulation.

3.5.10 Cooling.- The inverter shall be self-cooled by natural convection, conduction, and radiation.

3.5.11 Warmup time.- The inverter shall require no warmup time.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:29Z
Check the source to verify that this is the current version before use.

3.5.12 Vibration isolators.- Unless otherwise approved by the procuring activity, vibration isolators, if used, shall be of such design that the main support between the stationary and isolated member is of all metal contruction.

3.5.13 Maintainability.- The inverter shall have a minimum number of parts consistent with reliability. Its design, where practicable, shall permit easy assembly, disassembly, location of trouble sources, and maintenance with tools and equipment normally available commercially, by service maintenance personnel with a minimum of training.

3.6 Performance.-

3.6.1 Dielectric strength.- The electrical components of the inverter, except for such critical components as transistors and capacitors, shall withstand a potential of 1,500V rms commercial frequency applied for 1 minute between the terminals and the enclosure.

3.6.1.1 Capacitors.- Capacitors shall withstand twice the peak voltage to which they will be subjected during service, or 100V, whichever is greater, for 1 minute. If tantalum capacitors are used, they shall conform to table I and MIL-C-3965.

TABLE I. Tantalum capacitor voltages

DC surge voltage	Capacitor dc working voltage rating
0 - 1.5V	6V
1.5 - 6.0V	30V
6.0 -30.0V	75V
30.0 -50.0V	100V
Over -50.0V	2 times peak

3.6.2 Output voltage regulation.- The inverter shall maintain steady-state voltage limits specified on the applicable MS.

3.6.3 Output frequency regulation.- The inverter shall maintain steady-state frequency limits specified on the applicable MS.

3.6.4 Overload.- When the input voltage is set at 18V, the inverter shall be capable of delivering 150 percent of rated load for 1 minute at the minimum output voltage and the lagging power factor specified on the applicable MS.

3.6.5 Short circuit capacity.- When the input is set at 18V, the inverter shall be capable of delivering a minimum of 300 percent of rated current to a zero impedance line-to-ground short circuit and to carry this current for 5 seconds without sustaining damage therefrom. Provisions may be incorporated to limit this short circuit current during the 5-second interval to any practical desired values greater than 300 percent of rated current. The current limits shall be comprised of solid-static circuitry and shall not open the output circuit. After the short circuit has been removed, the inverter shall automatically return to normal operation without impairment of reliability or subsequent performance.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:29Z
Check the source to verify that this is the currentversion before use.

3.6.6 Efficiency.- The inverter shall provide the efficiency specified on the applicable MS.

3.6.7 Phase rotation (3-phase inverters).- For 3-phase inverters, phase rotation shall be in the order of A-B-C.

3.6.8 Phase unbalance (3-phase inverters).- The phase voltage unbalance of the inverter shall not exceed 2 percent under the following load conditions:

(a) No 3-phase load and a single-phase load equal to 2/9 of the 3-phase rating.

(b) A 3-phase load equal to 1/3 of the rating and a single-phase load equal to 2/9 of the 3-phase rating.

(c) A 3-phase load equal to 2/3 of the rating and a single-phase load equal to 1/9 of the 3-phase rating.

3.6.9 Output voltage modulation.- The voltage modulation, less the direct current input voltage modulation, shall not exceed 0.5 percent under conditions of no load, half load, and full load with lagging power factor as specified on the applicable MS.

3.6.10 Output frequency modulation.- The frequency modulation shall not exceed the values specified in MIL-STD-704.

3.6.11 Waveform.- The line-to-line crest factor shall not exceed the limits of 1.414 ±10 percent, and the observed change in the crest factor from no load to full load at any input voltage shall not exceed 10 percent. The total harmonic content and the individual harmonic content shall not exceed the values specified in MIL-STD-704.

3.6.12 Transient voltage.- Any transient output voltage shall not exceed the limits of MIL-STD-704 and shall recover to and remain within 10 percent of its steady-state condition within 0.1 second.

3.6.12.1 Transient output voltage.- The transient output voltage shall not exceed the values in MIL-STD-704. This requirement also shall apply to overvoltage spikes of microsecond duration. Moreover, during design, careful consideration shall be given to the protection of the inverter against any overvoltage spikes that may be internally generated.

3.6.13 Reduced input voltage.- The inverter shall be capable of carrying rated load when operated continuously at an input voltage of 18V dc.

3.6.14 Life.- The inverter shall be capable of operating for a minimum of 2,500 hours without maintenance under conditions of rated load, input of 27.5V dc, lagging power factor and high ambient temperature as specified on the applicable MS, and at approximately sea level altitude. As a design objective the unit shall have a life of at least 10,000 hours.

3.6.15 Motor starting.- Single- or 3-phase inverters shall provide full starting torque to single- or 3-phase induction motors, respectively, requiring an inrush current of 200 percent of inverter rated current at 0.5 lagging power factor.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:29Z
Check the source to verify that this is the currentversion before use.

3.6.16 Phase displacement angle (3-phase inverters).- The displacement between adjacent phases shall not exceed the limits specified in MIL-STD-704.

3.6.17 Endurance.- The inverter shall operate satisfactorily when subjected to 200 hours of intermittent operation consisting of 5 seconds with the switch on and 5 seconds with the switch off.

3.6.18 Environmental conditions.- The inverter shall provide electrical energy in flight vehicles under any of the following conditions or natural combination of conditions:

(a) Barometric pressures ranging from 30 inches Hg to the barometric pressure at the maximum altitude specified on the applicable MS.
(b) High ambient temperature-altitude conditions as specified on the applicable MS.
(c) Low temperature constant at -55° C at sea level.
(d) Relative humidity ranging up to 100 percent landing including conditions wherein condensation takes place on the equipment.
(e) Exposure to salt-sea atmosphere.
(f) Vibration incident to service use.
(g) Shock forces of 50g.
(h) Fungus growth as encountered in tropical climates.
(i) Sand and dust particles as encountered in desert areas.
(j) Explosive vapor within or surrounding the equipment.

3.6.19 Acceleration.- Voltage and frequency regulation shall not be affected when the inverter is subjected to an acceleration force of 10g along the major axes.

3.6.20 Operating position.- The inverter shall be capable of operating in any position.

3.6.21 Source impedance.- The inverter shall be capable of operating satisfactorily when connected to a 28V dc generator having a power rating of the inverter.

3.7 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of MIL-I-70327 shall govern changes in the manufacturer's part numbers.

3.8 Radio interference.- The inverter shall be designed to minimize the generation of radio interference. Enclosing case construction shall provide continuity of electrical shielding with a low radio frequency impedance path to ground and across all mechanical discontinuities. Conducted and radiated interference produced by operation of the inverter shall not exceed the requirements of MIL-STD-826.

3.9 Dimensions.- The overall dimensions shall be held to a minimum, and in no case shall they exceed those specified on the applicable MS. The mounting dimensions shall be in exact accordance with the applicable MS.

3.10 Weight.- The weight shall be held to a minimum and in no case shall it be in excess of that specified on the applicable MS.

3.11 Screw threads.- Screw threads shall be in accordance with MIL-S-7742.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:29Z
Check the source to verify that this is the current version before use.

3.12 Finishes and protective coatings.-

3.12.1 Aluminum alloy parts.- Aluminum alloy parts shall be anodically treated in accordance with MIL-A-8625. To provide adequate bonding, the aluminum oxide film formed by this treatment shall be removed from the actual contact area of all surfaces required to act as a path for electrical current and from the local areas under screws, nuts, etc., used for assembly or mounting purposes.

3.12.2 Magnesium alloy parts.- Magnesium alloy parts shall be surface treated in accordance with MIL-M-3171 to provide protection against corrosion.

3.12.3 Steel parts.- Steel parts shall be cadmium plated in accordance with type II or III of QQ-P-416, and of a class that is adequate to achieve the degree of protection required.

3.13 Identification of product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Workmanship.- The inverter, including all parts and accessories, shall be fabricated in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, radii of fillets, and marking of parts and assemblies; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; alignment of parts, and tightness of assembly screws and bolts.

3.14.1 Dimensional tolerances.- Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the inverter, they shall be held or limited accordingly.

3.14.2 Screw assemblies.- Assembly screws and bolts shall be tight. The word tight means that the screw or bolt cannot be appreciably tightened further without damage to the screw, bolt, or threads.

3.14.3 Riveting.- Riveting operations shall be carefully performed to ensure tightness and satisfactory heading of rivets.

3.14.4 Cleaning.- The inverter shall be thoroughly cleaned and loose, spattered, or excess solder, metal chips, and other foreign materials removed during and after final assembly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:29Z
Check the source to verify that this is the currentversion before use.

4.2 Classification of inspections.- The examining and testing of inverters shall be classified as follows:

(a)	Qualification inspections	(4.3)
(b)	Quality conformance inspections	(4.4)

4.3 Qualification inspections.-

4.3.1 Qualification samples.- The qualification samples shall consist of three inverters. Samples shall be identified with the manufacturer's own part number and any additional information required by the letter of authorization (see 6.3).

4.3.1.1 Drawings and instructions.- Together with the test samples, the manufacturer shall submit one reproducible copy of the outline and assembly drawing and two printed copies of simple instructions with illustrations and diagrams, if necessary, covering the installation of the inverter.

4.3.2 Qualification required.- Prior to actual procurement, the product which this specification covers shall pass the qualification tests specified herein. If the product is later modified in any way, the modified form shall be subjected to and shall pass the same qualification tests.

4.3.3 Inspections.- The qualification inspections shall consist of all the examinations and tests described under 4.6. The tests shall be conducted on the sample inverters as follows and essentially in the order listed:

(a) Sample No. 1

(1)	Examination of product	(4.6.1)
(2)	Dielectric strength	(4.6.2)
(3)	Phase rotation	(4.6.7)
(4)	Voltage and frequency regulation	(4.6.3)
(5)	Efficiency	(4.6.6)
(6)	Phase unbalance (3-phase inverters)	(4.6.8)
(7)	Modulation	(4.6.9)
(8)	Waveform	(4.6.10)
(9)	Radio interference	(4.6.21)
(10)	Transient voltage recovery	(4.6.11)
(11)	Reduced input voltage	(4.6.13)
(12)	Motor starting	(4.6.15)
(13)	Overload	(4.6.4)
(14)	Short circuit capacity	(4.6.5)
(15)	Factors affecting voltage and frequency regulation	(4.6.3)
(16)	Humidity	(4.6.18.3)
(17)	Disassembly inspection	(4.6.22)

(b) Sample No. 2

(1)	Examination of product	(4.6.1)
(2)	Dielectric strength	(4.6.2)
(3)	Phase rotation	(4.6.7)
(4)	Voltage and frequency regulation (room temperature only)	(4.6.3(a))

Source: https://assist.dla.mil -- Downloaded: 20260-20T11:29Z
Check the source to verify that this is the currentversion before use.

(5)	Efficiency	(4.6.6)
(6)	Modulation	(4.6.9)
(7)	Waveform	(4.6.10)
(8)	Phase displacement angle (3-phase inverters)	(4.6.16)
(9)	Factors affecting voltage and frequency regulation	(4.6.3)
(10)	Shock	(4.6.18.6)
(11)	Vibration	(4.6.18.5)
(12)	Fungus resistance	(4.6.18.7)
(13)	Salt spray	(4.6.18.4)
(14)	Explosion-proof	(4.6.18.9)
(15)	Radio interference	(4.6.21)
(16)	Endurance	(4.6.17)
(17)	Disassembly inspection	(4.6.22)

(c) Sample No. 3

(1)	Examination of product	(4.6.1)
(2)	Dielectric strength	(4.6.2)
(3)	Phase rotation	(4.6.7)
(4)	Voltage and frequency regulation (room temperature only)	(4.6.3(a))
(5)	Efficiency	(4.6.6)
(6)	Operating position	(4.6.20)
(7)	Modulation	(4.6.9)
(8)	Waveform	(4.6.10)
(9)	Input transient voltage protection	(4.6.12)
(10)	Voltage unbalance	(4.6.8.1)
(11)	Low temperature	(4.6.18.2)
(12)	Sand and dust	(4.6.18.8)
(13)	Reduced input voltage	(4.6.13)
(14)	Radio interference	(4.6.2.1)
(15)	Acceleration	(4.6.19)
(16)	Life	(4.6.14)
(17)	Disassembly inspection	(4.6.22)
(18)	Source impedance	(4.6.14.1)
(19)	Disassembly inspection	(4.6.22)

4.6.3.1 Qualification test report.- After completion of the qualification tests, the contractor shall prepare a qualification test report in accordance with MIL-STD-831 and furnish three complete copies to the procuring activity.

4.4 Quality conformance inspections.- The quality conformance inspections shall consist of the individual inspections.

4.4.1 Individual inspections.- Each inverter shall be subjected to the following inspections:

(a)	Examination of product	(4.6.1)
(b)	Dielectric strength	(4.6.2)
(c)	Voltage and frequency regulation (room temperature only)	(4.6.3(a))
(d)	Efficiency	(4.6.6)
(e)	Phase rotation (3-phase inverters)	(4.6.7)

Source: https://assist.dla.mil -- Downloaded: 20260-20T11:29Z
Check the source to verify that this is the current version before use.

4.5 Test conditions.-

4.5.1 Unless otherwise specified, tests shall be made under the following conditions:

(a) An ambient temperature of 25° ±15° C.
(b) At approximately sea level altitude.
(c) Input and output voltages measured at the terminals of the inverter.
(d) Lagging power factor loads as specified on the applicable MS.
(e) Input voltage of 27.5V dc.
(f) Inverter mounted in a normal upright position with mounting base in a horizontal plane.
(g) Balanced 3-phase, wye-connected loading.
(h) With a 400-amp, 28V dc aircraft generator supplying input power.

4.6 Inspection methods.-

4.6.1 Examination of product.- The inverter shall be inspected to determine compliance with the requirements of this specification and the applicable MS with respect to materials, workmanship, dimensions, weight, and marking. The results shall be recorded in the qualification test report.

4.6.2 Dielectric strength.- While the inverters are hot as a result of testing, a potential of 1,500V rms commercial frequency shall be applied for 1 minute between the metal enclosure and all the terminals shorted together. No breakdown of the insulation shall occur as a result of this test. If the preceding method of testing is not feasible, the dielectric tests may be conducted on the components prior to final assembly or with the critical components disconnected.

4.6.2.1 Capacitors.- Capacitors, except for tantalum capacitors, shall be subjected to and shall withstand for 1 minute a dc test voltage of 100V or twice the maximum peak voltage encountered during normal operating conditions, whichever is greater. Tantalum capacitors, if used, shall be subjected to the surge voltage test specified in MIL-C-3965.

4.6.3 Voltage and frequency regulation.- The regulated output voltage and frequency shall be measured under the following conditions. For each temperature condition, the voltage and frequency shall remain within the limits specified on the applicable MS.

(a) Voltage and frequency regulation at ambient temperature (25° ±15° C):

(1) At minimum specified input voltage, the load shall be varied from no load to full load in steps of one-fourth of rating. The load shall be suddenly removed and applied for each observation. Voltage and frequency readings shall be taken to accurately plot the voltage and frequency regulation at the minimum lagging and leading power factor loads, as specified on the applicable MS.

(2) At rated input voltage, the test specified in 4.6.3(a)(1) shall be repeated.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:29Z
Check the source to verify that this is the current version before use.

(4) At zero output current, the input voltage shall be gradually varied from minimum specified input voltage to maximum specified input voltage and returned. The output voltage shall not be adjusted and data need be recorded only at minimum and maximum input voltage conditions.

(5) At 50 percent rated output current, the test specified in 4.6.3(a)(4) shall be repeated.

(6) At 100 percent rated output current, the test specified in 4.6.3(a)(1) shall be repeated.

(b) Voltage and frequency regulation as specified under 4.6.3(a), except that the temperature shall be -55° C. Immediately prior to this test, the inverter shall be soaked for 6 hours at -55° C.

(c) Voltage and frequency regulation as specified under 4.6.3(a), except that the temperature shall be the high ambient temperature specified on the applicable MS. Immediately prior to this test, the inverter shall be soaked at the high temperature specified on the MS.

.4 Overload.- The inverter shall be tested to demonstrate its ability to deliver percent of rated load for 1 minute at the lagging power factor and the minimum put voltage specified on the applicable MS with the input voltage at 18V.

.5 Short circuit capacity.- Each phase of the inverter shall be tested to demonste its ability to deliver a minimum of 300 percent of rated output current to a o impedance line-to-ground short circuit for 5 seconds with the input voltage at . No damage shall occur to the inverter as a result of this test.

.6 Efficiency.- The power output and input shall be measured at full load and lagging power factor specified on the applicable MS. The efficiency as calculated this condition shall be not less than that specified on the MS.

.7 Phase rotation (3-phase inverters).- The phase rotation of the inverter shall determined. The phase rotation shall be in the order of A-B-C.

.8 Phase unbalance (3-phase inverters).- The maximum value of output voltage alance occurring during the tests specified under 4.6.8.1 shall not exceed 2 per-t. The percent of unbalance of the phase voltages shall be defined as 100 times maximum deviation of the phase voltage from the average of the 3-phase voltages ided by the average of the 3-phase voltages.

.8.1 Unbalanced loads.- The effects of unbalanced 3-phase loads on the balance the 3-phase voltage output of the inverter shall be determined at both minimum and imum input voltage as follows (single-phase loads shall be connected line-to-neutral wye-connected inverters and line-to-line on delta-connected inverters):

(a) With the inverter carrying no 3-phase load, a single-phase load of approximately 2/9 of the full 3-phase rating shall be connected successively to each phase.

Source: https://assist.dla.mil -- Downloaded: 202640-20T11:29Z
Check the source to verify that this is the currentversion before use.

(b) With the inverter carrying a balanced 3-phase load of 1/3 of rated load, a single-phase load of approximately 2/9 of the full 3-phase rating of the inverter shall be connected successively to each phase.

(c) With the inverter carrying a balanced 3-phase load of 2/3 of rated load, a single-phase load of approximately 1/9 of the full 3-phase rating of the inverter shall be connected successively to each phase.

4.6.8 2 Symmetry of construction.- For all balanced loads, irrespective of power factor, within the rating of the unit, the 3-phase output voltage shall not be unbalanced by more than 0.5 percent.

4.6.9 Modulation.-

4.6.9.1 Output voltage modulation.- The inverter shall be operated at rated input voltage and its output voltage modulation determined by means of a calibrated oscilloscope or other means satisfactory to the procuring activity. The modulation, less the input voltage modulation, shall not exceed 0.5 percent at no load, half load and full load at the lagging power factor as specified on the applicable MS and shall be computed according to the following formula:

$$\text{Percent modulation} = \frac{E_{max} - E_{min}}{E_{max} + E_{min}} \times 100$$

4.6.9.2 Input frequency modulation.- The inverter shall be measured under conditions of no load, half load, and full load and shall not exceed the values specified in MIL-STD-704. Measurements of frequency modulation shall be made by a method approved by the procuring activity.

4.6.10 Waveform.- The crest factor and harmonic content of the output voltage shall be determined under the following conditions:

(a) At minimum input voltage for no load, half load, and full load.

(b) Repeat for full load at the lagging power factor specified on the applicable MS.

(c) Repeat for full load at the leading power factor specified on the applicable MS.

(d) Repeat (a), (b), and (c) for rated input voltage.

(e) Repeat (a), (b), and (c) for maximum input voltage.

4.6.10.1 The line-to-line crest factor obtained during this test shall not exceed 1.414 ±10 percent, and the observed change in crest factor from no load to full load at any input voltage shall not exceed 10 percent. The crest factor shall be determined by means of an oscillograph or peak and rms voltmeters. When voltmeters are used, the peak reading instrument shall be connected successively for each of the two possible polarities, and the crest factor shall be the ratio of the average of the two peak values to the reading of the rms voltmeter. The meters shall be of approved types. The total harmonic content and the individual harmonic content shall not exceed the values specified in MIL-STD-704. The harmonic content shall be determined by an approved analyzer.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:29Z
Check the source to verify that this is the current version before use.

.11 Transient voltage.- A recording oscillograph shall be used to measure the put voltage transient when full-rated load is applied and removed and vice versa. transient output voltage shall not exceed limits 2 and 3 of MIL-STD-704 and shall over to and remain within 10 percent of the steady-state condition within 0.1 second n application or removal of rated load. The unit shall not develop sustained illations under any loading condition.

.12 Input transient voltage protection.- Step function voltages shall be applied appropriate periods, as shown in table II. Prior to starting the test the inverter ll have been operating at rated load and maximum ambient sea level temperature cified on the applicable MS for a period sufficient for the inverter to have attain- maximum working temperature. The inverter shall be subjected to four tests and h test shall consist of one application of:

(1) Nominal voltage for the duration of period "A".
(2) Voltage "V" for the duration of period "B".
(3) Nominal voltage for the duration of the period "C".

re shall be no flash over and no insulation break-down. At the conclusion of this t the voltage and frequency regulation shall be within the limits specified on the licable MS.

TABLE II. Step-function voltages

Test No.	Period "A"		Period "B"		Period "C"		Number of applications
	Minutes	Volts	Seconds	Value "V" volts	Minutes	Volts	
1	7.5	28	0.04	80	7.5	28	1
2	7.5	28	0.13	68	7.5	28	1
3	7.5	28	5.0	36	7.5	28	1
4	7.5	28	10^{-5}	200	7.5	28	1

.13 Reduced input voltage.- The inverter shall be tested to demonstrate its lity to carry full-rated load within the output voltage limits specified on the licable MS with the input voltage varied from 18 to 29V in increments of 1V.

.14 Life.- At rated input voltage and the high temperature specified on the licable MS, the inverter shall be operated at sea level with unity power factor l load for 500 hours. The inverter shall be subjected to a voltage and frequency ulation test as specified in 4.6.3(c) at the beginning and at the end of this test. voltage and frequency regulation shall be within the limits specified on the appli- le MS. This test shall be repeated for 2,000 hours at room temperature (25 ±15° C). malfunction of the inverter during this test shall be cause for rejection.

.14.1 Source of impedance.- One hundred twenty five hours of the 500 hour test all be conducted using an aircraft 28V dc generator having a current rating of not e than twice the input current rating of the inverter or 25 amps., whichever is eater, for supplying power to the inverter. Performance shall be satisfactory and ere shall be no failure of input capacitors or other components of the inverter as result of this test.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:29Z
Check the source to verify that this is the currentversion before use.

4.6.15 Motor starting.- The inverter shall demonstrate its capability to start an induction motor load varying from 200 percent of inverter rated VA, 0.5 lagging power factor at initial inrush to 100 percent of inverter rated VA, 0.85 lagging power factor at normal steady-state operation. Single-phase inverters shall be tested with single phase motors and 3-phase inverters with 3-phase motors. A dummy load, which simulates the motor characteristics may be utilized in lieu of the motor.

4.6.16 Phase displacement angle (3-phase inverters).- The displacement between adjacent phases shall be determined at no load, half load, and full load. The 3-phase load shall be balanced for this test. The displacement between adjacent phases shall be within the limits of 120 ±1.5 degrees as specified in MIL-STD-704.

4.6.17 Endurance.- The inverter shall be operated for 200 hours during which time full-rated load at minimum lagging power factor shall be switched on for 5 seconds and off for 5 seconds. Sufficient readings shall be taken to ensure compliance with this test. Following this test, the unit shall pass the tests specified in 4.6.3(a)(2). During this test, readings shall be taken at least every 12 hours to demonstrate that the output voltage and frequency does not drift outside the limits specified on the applicable MS.

4.6.18 Environmental.-

4.6.18.1 Factors affecting voltage and frequency regulation.- Prior to each of the following tests (a) through (c), the voltage and frequency shall be at a value termed "standard." For the purposes of this test, the voltage and frequency "standard" shall be a value that lies between the limits specified on the applicable MS and that which has been determined after voltage, frequency, and temperature stabilize. The maximum deviation of voltage and frequency in both positive and negative directions from the standard shall be recorded for each test.

(a) The ambient temperature shall be decreased from 25° ±15° to -55° C, increased to the maximum ambient temperature specified on the applicable MS, and then checked at 25° ±15° C. Each temperature shall be held for at least 1 hour or until the regulated voltage and frequency stabilize as indicated by readings taken at 10-minute intervals. The ability of the inverter to continuously deliver rated output at ambients between -55° C and the maximum ambient temperature specified on the applicable MS shall be demonstrated, and the value of the voltage and frequency recorded.

(b) The load power factor shall be decreased from maximum to both the minimum lagging and the minimum leading value specified on the applicable MS. Each load power factor shall be held until the regulated voltage and frequency stabilize.

(c) The pressure altitude shall be varied from approximately sea level to conditions approximately the maximum altitude specified on the applicable MS. Tests shall be made at intermediate altitudes at the option of the procuring activity. Each altitude shall be held until the regulated voltage and frequency stabilize. The ambient temperature at each altitude shall be within ±5° C of the maximum on MS33543 for the range indicated on the applicable MS.

Source: https://assist.dla.mil -- Downloaded: 2026-01-20 11:29Z
Check the source to verify that this is the current version before use.

.6.18.1.1 For both voltage and frequency, the five values of maximum positive deviations recorded from tests (a) through (c) shall be added together and the five values f maximum negative deviations shall be added together. To these sums add, respectvely, the maximum positive and the maximum negatives deviations owing to changes in oad and input voltage at room ambient that were recorded during the tests specified n 4.6.3a(1), (2), and (3). The total positive and the total negative deviations thus btained shall be added algebraically to the standard values of voltage and frequency eparately, and the inverter shall be considered acceptable if the voltage and freuency values thus obtained remain within the limits specified on the applicable MS.

.6.18.2 Low temperature.- The inverter shall be subjected to low temperature test rocedure II of MIL-E-5272, except that the temperature for the entire test shall be 55° C. During the final hour of the test the inverter shall be energized and subected to the voltage and frequency regulation test specified in 4.63(a)(1), (2), and 3).

.6.18.3 Humidity.- The inverter shall be subjected to humidity test Procedure I f MIL-E-5272. At the end of this test, the inverter shall be immediately subjected o the tests specified in 4.6.3(a)(1), (2), and (3).

.6.18.4 Salt spray.- The inverter shall be subjected to salt spray test Procedure I f MIL-E-5272 for 50 hours. At the end of this test, the inverter shall be washed nd dried for 15 to 20 hours and shall then be subjected to the tests specified in .6.3(a)(1), (2), and (3). There shall be no evidence of excessive corrosion.

.6.18.5 Vibration.- The inverter shall be subjected to vibration test Procedure XII f MIL-E-5272, except that in lieu of the 71° C requirement of the vibration test chedule, the temperature shall be the high temperature specified on the applicable S. The inverter shall be operated at rated load and the output voltage and frequency hall remain within the specified limits throughout this test.

.6.18.6 Shock.- The inverter shall be subjected to shock test Procedure V of IL-E-5272, except that the impact shock shall be 50g. Damage or failure of any part f the inverter during this test shall be cause for rejection. Following these tests, he inverter shall satisfactorily withstand the tests specified in 4.6.3(a)(1), (2), nd (3).

.6.18.7 Fungus resistance.- The inverter shall be subjected to fungus resistance est Procedure I of MIL-E-5272. After completion of this test, the inverter shall how no appreciable growth of fungi and shall satisfactorily withstand the tests specified in 4.6.3(a)(1), (2), and (3).

.6.18.8 Sand and dust.- The inverter shall be subjected to sand and dust test rocedure I of MIL-E-5272. After completion of this test, the inverter shall operate atisfactorily and shall satisfactorily withstand the tests specified in 4.6.3(a)(1), 2), and (3).

.6.18.9 Explosion proof.- The inverter shall be subjected to explosion-proof test rocedure III of MIL-E-5272.

.6.19 Acceleration.- The inverter shall be mounted on a centrifuge and subjected o an acceleration of 10g along each of the major axes. The acceleration shall prouce no change in the outvoltage or frequency.

Source: https://assist.dla.mil -- Downloaded: 202610-20T11:29Z
Check the source to verify that this is the currentversion before use.

4.6.20 Operating position.- During the course of other tests, it shall be ascertained that the operation of the inverter in any position does not adversely affect power input, heating, and other operating characteristics for all inverters. Full operating life shall be attainable in any position.

4.6.21 Radio interference.- At no load, half load, and full load, both conducted and radiated radio interference shall be measured using the test procedures and any applicable test instruments specified in MIL-STD-826. Inverters exceeding the interference limits specified in MIL-STD-826 shall be rejected.

4.6.22 Disassembly inspection.- At the conclusion of all tests, the inverter shall be disassembled and inspected for excessive wear and defects.

4.7 Inspection of preservation, packaging, and packing.- The preservation, packaging, and packing shall be examined to determine conformance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking.- The preservation, packaging, packing, and marking shall be in accordance with MIL-P-7936. Level A preservation shall be in accordance with Method II of MIL-P-116.

5.2 Shipment marking nomenclature.- The shipment marking nomenclature shall be the nomenclature of the item shown on the applicable MS.

6. NOTES

6.1 Intended use.- The inverter covered by this specification is intended for use in converting electrical power in flight vehicles from 28V dc to ac power at 400 cps. It is intended for use primarily in flight vehicle applications that exceed the temperature and altitude requirements of present equipment.

6.2 Ordering data.- Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Data requirements (see 3.2).
(c) Selection of the applicable levels of preservation, packaging, and packing (see 5.1).

6.3 Qualification.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in the applicable Qualified Products List, whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Systems Engineering Group (SEG), Attn: SEGPE, Wright-Patterson Air Force Base, Ohio, and information pertaining to qualification of products may be obtained from that activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:29Z
Check the source to verify that this is the current version before use.

ustodians:
Army - MO
Air Force - (11)

eviewer activities:
Army - MO
Air Force - (11)

Preparing activity:
Air Force - (11)

Project No. 6125-0082

eview/user information is current as of the date of this document. For future oordination of changes to this document, draft circulation should be based on the nformation in the current Federal Supply Classification Listing of DoD Standardization Documents.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T11:29Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved n the use of the specification in procurement of products for ultimate use by the Depart- ent of Defense. This sheet is provided for obtaining information on the use of this pecification which will insure that suitable products can be procured with a minimum nount of delay and at the least cost. Comments and the return of this form will be ppreciated. Fold on lines on reverse side, staple in corner, and send to preparing ctivity.

PECIFICATION MIL-I-27273A Inverter, Power, Static, General Specification for

RGANIZATION | CITY AND STATE

ONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

ATERIAL PROCURED UNDER A

☐ Direct Government Contract ☐ Subcontract

. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

. REMARKS (Attach any pertinent data which may be of use in improving this specifica- tion. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.)

UBMITTED BY (Printed or typed name and activity) DATE

) Form 1426

Source: https://assist.dla.mil -- Downloaded: 20260-20T11:29Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commander, SEC

Attn: SEPS

Wright-Patterson AFB, Ohio 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T11:29Z
Check the source to verify that this is the current version before use.